

Fully Online Retrieval Adaptation

Executive Summary

29 August 2026

Current evidence

No current-code cluster experiment has completed and been analyzed. Consequently there is no accuracy conclusion, selected ablation value, or claim that online adaptation improves a frozen backbone.

Implemented protocol

The project precomputes T0, T1+T2, and T3 indexes before extraction. Each query retrieves from mature origins only, and ridge or gate fitting uses exactly 10 already matured dates for the query user by default. Retrieval considers at most 10,000 candidate windows across users. The default scopes remain all-user raw Euclidean retrieval, same-user fitting, and non-fixed datastore and training sets; explicit fixed variants and all-user fitting remain available.

At every forecast, the oldest 80% of the causal fitting window trains candidate ridges and the newest 20% selects $\alpha \in \{10^{-1}, 10^{-2}, 10^{-3}\}$ and $K \in \{1, 5, 10, 15\}$; the chosen pair is refitted on the complete fitting window. CSV NaNs are filled with zero after aggregation unless `missing_values=error`; infinite values are rejected.

Extraction is persistent and independently manifested, so matched adaptation parameters can reuse matching features. Reports require identical query dates. The fixed-protocol ablation crosses explicit fixed-datastore and fixed-training-set booleans while retaining the same T3 grid.

Diagnostic cluster evidence

Job	Observed cause and interpretation
2964440	Compact virtual-array context was truncated below the required 168-step lookback; fixed-protocol accuracy was never evaluated.
2964441	TS-RAG tokenizer tensors were on CUDA while bucket boundaries remained on CPU; TS-RAG accuracy was never evaluated.
2964508	All 12 fixed remainder adaptations completed; reporting failed because extraction hashes split equivalent vanilla model labels.
2964507	All 16 extractions and the first ridge adaptation completed; the source-adapted TS-RAG ARM then imported a missing class from the thin official Chronos-Bolt adapter.
2964505 / 2964506	Both anchors reached their final Solar adaptation with empty stderr, but their old logs have no terminal marker, so final scheduler states are not independently verified locally.
2964439	Older online run belongs to the superseded runtime contract and is not eligible even if it later terminates.
2955394 / 2913487	Obsolete running-manifest and quota-bound execution paths; diagnostic only.

The current code slices compact contexts correctly, completes tokenizer/bucket operations on CPU before device transfer, isolates the pinned TS-RAG Chronos-Bolt core, collapses equivalent vanilla sources to one readable model label, and emits explicit terminal records. These implementation failures do not select an adaptation method or hyperparameter.

Next evidence

Deploy the current code. Resume priority TS-RAG with only the adaptation and table stages, reusing its 16 extractions and completed first ridge result. Regenerate the fixed-protocol table from the already completed adaptations. The broader Electricity, Solar, Traffic, and later full-profile grid should follow only after these current-code smoke paths are valid.

Interpretation limit

Published TS-RAG and Cross-RAG MSE values remain contextual. The dedicated TS-RAG workflow matches its fixed same-variable retrieval scope and split boundaries, but preprocessing and metric scales must be checked before local and published results are combined.